

Assignment 1

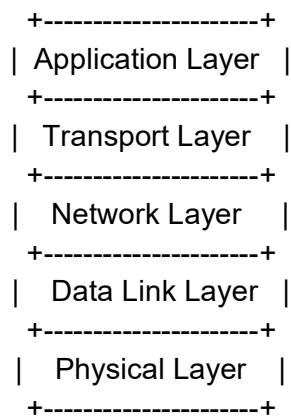
PCCST501 – Computer Networks

How the Internet Delivers a Web Page

Website Chosen: Amazon (<https://www.amazon.com>)

1. Draw the TCP/IP Protocol Stack and Explain the Role of Each Layer

The TCP/IP protocol suite is a five-layer model that lets devices talk to each other across the internet. Each layer has its own job and helps the layer right above it.



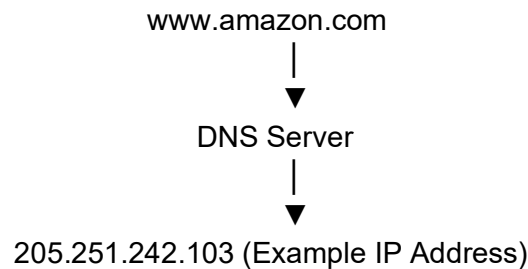
- **Application Layer:** This is the top layer that users and apps interact with directly. It handles protocols like HTTP for web browsing, SMTP for emails, and DNS.
- **Transport Layer:** This layer gets data from the application layer, chops it into smaller pieces called segments, and makes sure it gets delivered properly. It mostly uses TCP for reliable delivery or UDP for speed.
- **Network Layer:** This layer handles moving data between different networks. It uses IP (Internet Protocol) addresses to route packets through routers to their final destination.
- **Data Link Layer:** This layer moves data across the local network link. It uses MAC addresses to frame the data and prep it for the physical wire or wireless signal.
- **Physical Layer:** This is the actual hardware layer. It turns the data into electrical, optical, or radio signals and sends them over cables or Wi-Fi.

2. Explain the Role of DNS in Converting the Domain Name into an IP Address

When you type [\[www.amazon.com\]\(https://www.amazon.com\)](https://www.amazon.com) into your browser, the computer doesn't know where to go because it only understands numbers, not words. The internet relies on IP addresses to find web servers.

1. Your browser asks a Domain Name System (DNS) server for help. DNS works on the application layer and acts like the phonebook of the internet.

2. The DNS server looks up the human-friendly name ([\[www.amazon.com\]\(https://www.amazon.com\)](https://www.amazon.com)) and translates it into its actual IP address (for example, [205.251.242.103](https://www.amazon.com)).
3. Once the browser gets this IP address back, it finally knows exactly where Amazon's servers are located and can connect to them.



3. Describe How HTTP Is Used to Request and Receive the Webpage

HTTP (HyperText Transfer Protocol) is the main rulebook for fetching web pages. It works on a simple client-and-server system. Here, your web browser is the client, and Amazon's computer is the server.

Here is how the conversation happens:

1. Your browser sends an HTTP request message to the Amazon server.
2. This request usually contains a GET command, asking the server to hand over the homepage.
3. The Amazon server reads the request and grabs the necessary files.
4. The server then replies with an HTTP response, which includes the code for the webpage.
5. Your browser reads the response and paints the Amazon homepage on your screen.

An HTTP request usually includes a request line, headers (extra info), a blank line, and sometimes a body. An HTTP response includes a status line (like "200 OK" if it worked), headers, a blank line, and the actual body (the webpage data).

4. Explain How TCP Provides Reliable Communication for HTTP

TCP (Transmission Control Protocol) is the backbone that makes sure HTTP works smoothly without losing pieces of the webpage.

- **Reliability:** TCP checks that every single packet of data actually made it to your computer.
- **Error Control:** If a packet gets lost or scrambled along the way, TCP spots the mistake and asks the sender to send that specific piece again.
- **Flow Control:** It makes sure the sender doesn't overwhelm the receiver by sending data faster than the computer can process it.

- **Congestion Control:** If the overall internet connection is really busy and clogged up, TCP slows down the sending speed to help clear the traffic jam.

5. Identify Whether the Application Follows Client–Server or Peer-to-Peer Architecture and Justify Your Answer

Amazon uses a **Client-Server** architecture.

In this setup, your laptop or phone's web browser is the "client." It reaches out to ask for data. Amazon operates massive web "servers" that store all the product pages, images, and user accounts. The server's only job is to wait for requests from clients and serve the data back to them. This makes sense for Amazon because a central server needs to securely manage millions of products and handle thousands of customers shopping at the exact same time.

6. Compare HTTP and FTP

Feature	HTTP (HyperText Transfer Protocol)	FTP (File Transfer Protocol)
Main Use	Browsing web pages on the internet.	Uploading and downloading files between computers.
Connections	Uses one TCP connection for the whole conversation.	Needs two TCP connections: one for sending commands and one for sending the actual data.
Commands	Uses commands like GET (to view a page) and POST (to submit a form).	Uses commands like STOR (upload), RETR (download), and DELE (delete).
Data Types	Usually moves HTML text, images, and website scripts.	Moves all sorts of files, folders, and large documents.

7. Explain How Cookies Improve User Experience with One Real-World Example

Cookies are tiny text files that a web server asks your browser to hold onto. When you visit a site, it gives you a cookie. The next time you click something on that site, your browser sends

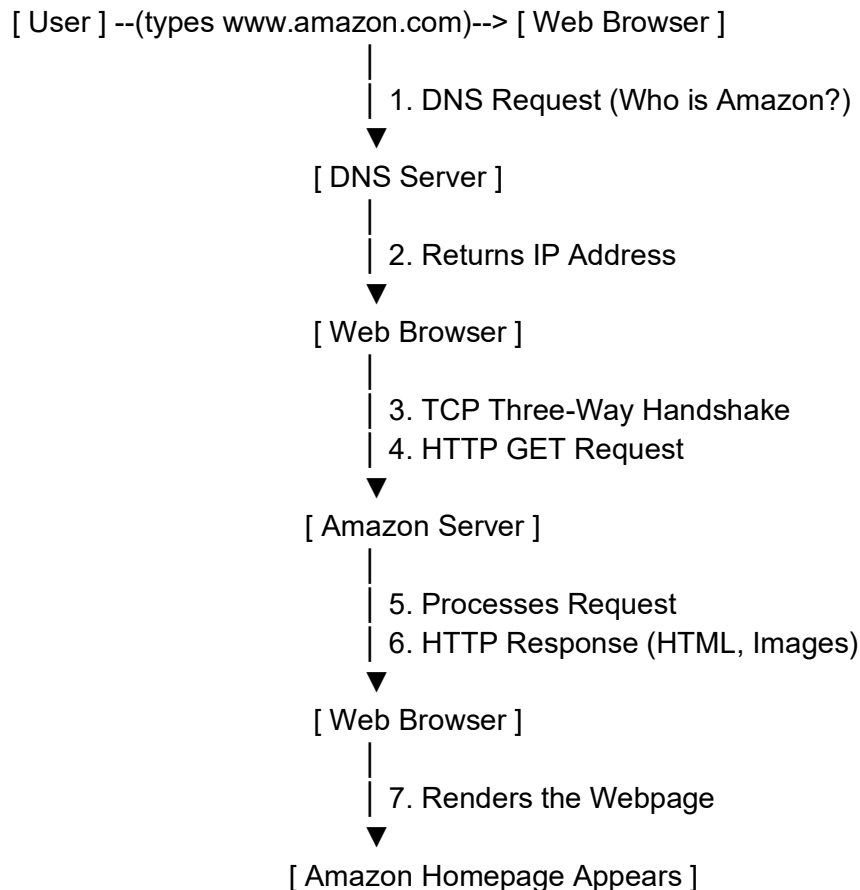
the cookie back. This lets the website "remember" who you are, so you don't have to keep re-entering the same information.

Real-World Example:

Imagine you are shopping on Amazon. You don't have an account yet, but you click "Add to Cart" on a pair of headphones. Amazon's server creates a temporary shopping cart for you and sends a cookie (let's say, ID #9876) to your browser. You then go look at phone cases and click "Add to Cart" again. Your browser hands ID #9876 back to the server. Because of that cookie, Amazon knows it's still you, and puts the phone case in the exact same cart as the headphones, instead of losing your stuff or starting a new cart.

8. Flow Diagram Showing the Complete Communication

Plaintext



9. Write a brief conclusion (100–150 words) describing the importance of application-layer protocols in everyday Internet usage.

Application-layer protocols are incredibly important because they are the invisible tools that make the internet actually usable for regular people. Without them, we wouldn't have the modern web. DNS saves us from having to memorize long strings of IP address numbers just to check the news or buy something online. HTTP acts as the universal language that lets our

browsers pull down text, images, and videos from servers across the globe instantly. Meanwhile, tools like cookies work in the background to remember our logins and shopping carts, creating a smooth and personalized experience. Together with the lower layers of the TCP/IP stack, these application protocols organize the chaos of network traffic into the easy, everyday services we rely on for school, entertainment, and communication.